Algebraik ifodalarni soddalashtirish

$4k + 5k$ ni $9k$ deb yozsa bo'ladi — bu **soddalashtirish**.

1-QISM

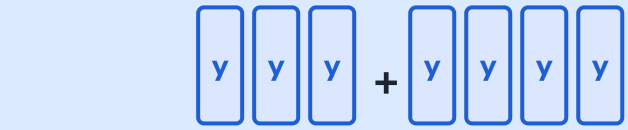
Mavzu

ESLAB OLAYLIK

- $4n$ — bu 4 ta n: $n + n + n + n = 4 \cdot n = 4n$.
- $1 \cdot n$ qisqacha n deb yoziladi. Demak, yolg'iz n — bu bitta n, ya'ni $1n$.

USUL · QO'SHISH

$3y + 4y$: 3 ta y va yana 4 ta y → jami $3 + 4 = 7$ ta y → $7y$.





Men avval qatorlarni sanadim — shunisi tezroq ekan.

1-MISOL · KO'CHATLAR

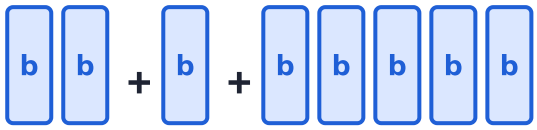
Bog'da 4 qator olma va 5 qator o'rik o'tqazildi, har qatorda k tadan ko'chat. Hammasi bo'lib nechta ko'chat bor?

1. Avval qatorlarni qo'shamiz: $4 + 5 = 9$ qator. Ko'chatlar: $9 \cdot k = 9k$ tup.
2. Boshqa yo'l: olma $4k$ tup, o'rik $5k$ tup. $4k + 5k = 9k$ tup.

Javob bir xil.

2-MISOL

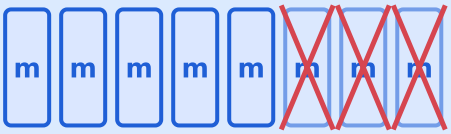
Qisqaroq yozing: $2b + b + 5b$.



1. $2b + 1b + 5b$.
2. $2 + 1 + 5 = 8$ ta b: $8b$.

USUL · AYIRISH

$8m - 3m$: 8 ta m dan 3 tasini olib tashlaymiz → $8 - 3 = 5$ ta m qoladi → $5m$.



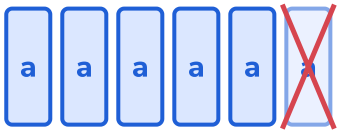
1-MISOL · POYEZD

Poyezdning har vagonida c ta o'rin bor. Poyezd 7 vagondan iborat edi, bekatda 3 vagon ajratib qoldirildi. Poyezdda nechta o'rin qoldi?

1. Avval $7c$ ta o'rin bor edi, $3c$ tasi ajratildi.
2. $7c - 3c = 4c$ ta o'rin qoldi.

2-MISOL

Soddalashtiring: $6a - a$.



1. $6a - 1a = 5a$.

Hammasi olib tashlansa, hech narsa qolmaydi: $6a - 6a = 0$.

USUL · HARFLI HADLAR VA SONLAR

$7r + 6 - 3r - 2$

$7r$, $3r$ kabi harfli qo'shiluvchilar — **harfli hadlar**. $7r$ bilan $3r$ ni yonma-yon qo'yamiz, 6 bilan 2 ni ham. Oldidagi minus yoki plyus ham birga ko'chiriladi:

$$7r - 3r + 6 - 2 \to 7r - 3r = 4r,\ 6 - 2 = 4 \to 4r + 4$$

1-MISOL

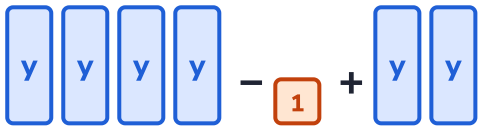
Soddalashtiring: $8 + 5x - 3 + x$.



1. Sonlar: $8 - 3 = 5$.
2. Harflilar: $5x + x = 5x + 1x = 6x$.
3. Javobda avval harfli had yoziladi: $6x + 5$.

2-MISOL

Ifodani qisqartiring: $4y - 1 + 2y$.



1. 1 oldidagi minus bilan birga oxirga ko'chadi: $4y + 2y - 1$.
2. $4y + 2y = 6y$. Son 1 ni $6y$ bilan birlashtirib bo'lmaydi: $6y - 1$.

2-QISM

Misollar ishlash

1-MISOL

Soddalashtiring: $10k - 3k + 4k - k$.

1. $10k - 3k = 7k$.
2. $7k + 4k = 11k$.
3. $11k - k = 11k - 1k = 10k$.

2-MISOL

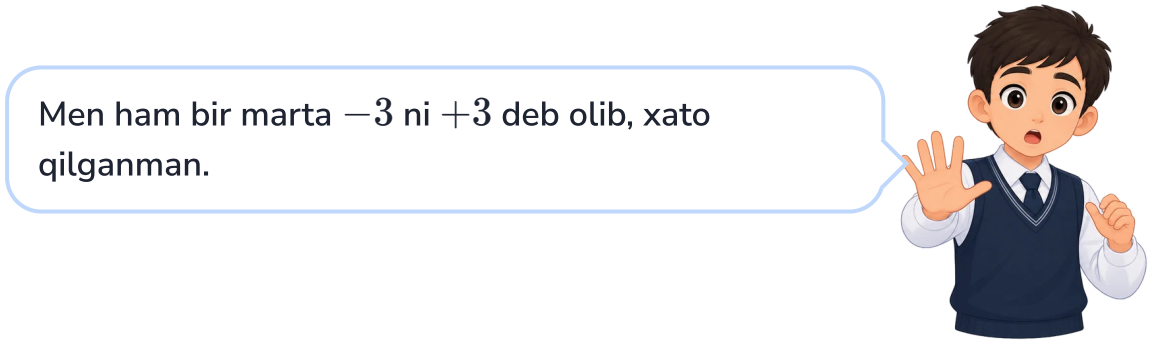
Qisqaroq yozing: $12z - 5 - 7z + 9$.

1. Harfli hadlar: $12z$ va $-7z$. Sonlar: -5 va $+9$.
2. Yonma-yon yozamiz: $12z - 7z + 9 - 5$.
3. $12z - 7z = 5z$.
4. $9 - 5 = 4$.
5. Javob: $5z + 4$.

3-MISOL

$\square$ o'rniga nima yoziladi? a) $14d = 9d + \square$; b) $\square - 3m = 8m$.

1. a) 9 ta d ga yana $14 - 9 = 5$ ta d qo'shilsa, 14 ta bo'ladi: $\square = 5d$.
2. Tekshiramiz: $9d + 5d = 14d$.
3. b) 3 ta m olib tashlangach, 8 ta qoldi. Avval $8 + 3 = 11$ ta edi: $\square = 11m$.



DIQQAT!

1. ✗ $5b + 5 = 10b$ → ✓ $5b + 5$ **shundayligicha qoladi**
2. ✗ $7a - a = 7a$ → ✓ $7a - 1a = 6a$
3. ✗ $8y - 3 - 2y + 7 = 6y + 10$ → ✓ $8y - 2y + 7 - 3 = 6y + 4$
4. ✗ $9p - 9p = p$ → ✓ $9p - 9p = 0$
5. ✗ $6n - 2n + 3n = 6n - 5n = n$ → ✓ $4n + 3n = 7n$

3-QISM

Mustahkamlash

1. Qisqaroq yozing. a) $d + d + d + d + d + d$; b) $w + w + w + w - w - w$; c) $3h + h + h$.
2. Soddalashtiring. a) $9c + 6c$; b) $17v - 8v$; c) $e + 12e$; d) $13x - x$.
3. Bitta had ko'rinishida yozing. a) $4g + 7g - 5g$; b) $15s - 6s - 2s$; c) $2z - z + 9z$; d) $11b - 3b + 4b - 7b$.
4. Plitkalarga qarab ifodani yozing va soddalashtiring.



5. Harfli hadlarni va sonlarni alohida hisoblang. a) $5k + 8 + 2k$; b) $3 + 10f - 1$.
6. Soddalashtiring. a) $12x + 7 - 5x - 3$; b) $18 + 5u - 7 + 3u$.
7. Ifodani qisqartiring: $14y - 5 - 6y + 9$.
8. Ifoda nimaga teng: $7p + 11 - 7p - 2$?
9. $\square$ o'rniga nima yoziladi? a) $16a = 11a + \square$; b) $\square - 6b = 2b$; c) $9c = \square \cdot c$.
10. **Xatoni toping.** Lola shunday yozdi: $10m - 4 - 3m + 9 = 7m - 13$. Qayerda xato? To'g'ri javobni toping.
11. Qaysi ifoda qolganlariga teng emas? a) $4x + 5 + 2x - 3$; b) $9x + 2 - 3x$; c) $3x + 8 + 3x - 6$; d) $7x + 2 - x + 2$.

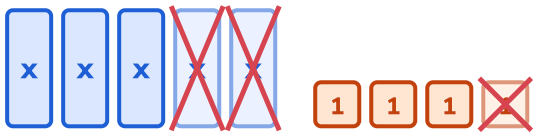
Matnli masalalar

1. Kinozalda 14 qator o'rindiq bor, har qatorda k ta o'rindiq. Ta'mir uchun oldingi 2 qator va oxirgi 1 qator yopib qo'yildi. Tomoshabinlar uchun nechta o'rindiq qoldi?
2. Novvoyxonada ertalab 10 patnis somsa pishirildi, har patnisda n ta somsa. Kechqurun 3 patnis somsa sotilmay qoldi. Nechta somsa sotilgan?
3. Bir pachkada p ta daftar bor. Maktab omboriga 6 pachka va yana 9 ta alohida daftar keltirildi. Sinflarga 2 pachka va 4 ta daftar tarqatildi. Omborda nechta daftar qoldi?
4. To'g'ri to'rtburchakning bo'yi $4b$ cm, eni b cm. Uning perimetrini (perimetr — barcha tomonlar yig'indisi) soddalashtirilgan ko'rinishda yozing.

4-QISM

Mustaqil ish

1. Qisqaroq yozing. a) $8q + 13q - 6q$; b) $30r - r - 14r$.
2. Soddalashtiring: $20k - 7 + 4k - 10k + 12$.
3. Rasmda ayrilgan plitkalar chizib qo'yilgan. Rasmga mos ifodani yozing va soddalashtiring.



4. $\square$ o'rniga nima yoziladi? a) $9t + \square + 6 = 13t + 6$; b) $\square + 2 - 5y = 3y + 2$.
5. $11y + 7 - 4y - \square$ ifodada y lar umuman qolmasligi uchun $\square$ o'rniga qanday harfli had yozish kerak? Shunda ifoda nimaga teng bo'ladi?
6. Kutubxonaga 12 to'plam kitob keldi, har to'plamda c ta kitob. 5 to'plam 6-A sinfga, 4 to'plam 6-B sinfga berildi. Kutubxonada nechta kitob qoldi?

5-QISM

Javoblar

Mustahkamlash

1-masala

a) 6 ta d: $6d$; b) $4 - 2 = 2$ ta w: $2w$; c) $3 + 1 + 1 = 5$, javob $5h$.

2-masala

a) $15c$; b) $9v$; c) $1 + 12 = 13$, javob $13e$; d) $13 - 1 = 12$, javob $12x$.

3-masala

a) $11g - 5g = 6g$; b) $9s - 2s = 7s$; c) $z + 9z = 10z$; d) $8b + 4b - 7b = 12b - 7b = 5b$.

4-masala

$4z + 2 + z + 3 = 4z + z + 2 + 3 = 5z + 5$.

5-masala

a) $5k + 2k + 8 = 7k + 8$; b) sonlar $3 - 1 = 2$, javob $10f + 2$.

6-masala

a) $12x - 5x + 7 - 3 = 7x + 4$; b) $5u + 3u + 18 - 7 = 8u + 11$.

7-masala

$14y - 6y + 9 - 5 = 8y + 4$.

8-masala

$7p - 7p = 0$, $11 - 2 = 9$. Javob: 9.

9-masala

a) $16 - 11 = 5$, javob $5a$; b) $2 + 6 = 8$, javob $8b$; c) 9.

10-masala

Lola $+9$ o'rniga -9 deb hisoblagan. To'g'risi: $10m - 3m + 9 - 4 = 7m + 5$.

11-masala

a), b), c) — $6x + 2$; d) $6x + 4$. Javob: d).

Matnli masalalar

1-masala

$14k - 2k - k = 12k - 1k = 11k$ ta o'rindiq.

2-masala

$10n - 3n = 7n$ ta somsa.

3-masala

$6p + 9 - 2p - 4 = 6p - 2p + 9 - 4 = 4p + 5$ ta daftar.

4-masala

$4b + b + 4b + b = 10b$ cm.

Mustaqil ish

1-masala

a) $21q - 6q = 15q$; b) $29r - 14r = 15r$.

2-masala

Harflilar: $20k + 4k - 10k = 14k$. Sonlar: $12 - 7 = 5$. Javob $14k + 5$.

3-masala

$5x + 4 - 2x - 1 = 5x - 2x + 4 - 1 = 3x + 3$.

4-masala

a) $13 - 9 = 4$, javob $4t$; b) $3 + 5 = 8$, javob $8y$ (tekshiruv: $8y + 2 - 5y = 3y + 2$).

5-masala

$11y - 4y = 7y$. y lar qolmasligi uchun yana $7y$ ayriladi: $\square = 7y$. Ifoda 7 ga teng.

6-masala

$12c - 5c - 4c = 3c$ ta kitob.